A-TMLE for Data Fusion Analyses (DRAFT)

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Agenda

1. RCT-RWD integration: motivations, examples, challenges
2. Estimator: Adaptive TMLE
3. Simulation studies
4. Case study: DEVOTE trial + Optum
5. Concluding remarks

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Augmenting RCT with external RWD

- **Enhance statistical power:**
  - RCTs often have limited time-frame, study of secondary endpoints/post-marketing safety outcomes may lack power;
  - E.g., testing for sub-components of a composite outcome;
- **Improve generalizability of trial findings:**
  - RCTs may have low external validity, external RWD includes a broader and more diverse trial-eligible patient population [1];
  - Extend findings to a target population that coincides with (or is a subset of) the trial-eligible population [2];
- **Transportation of trial findings:**
  - Extend effects to populations different from the trial [2];
  - E.g., trial conducted in Europe, transporting the estimated effects to the US population.
- ...

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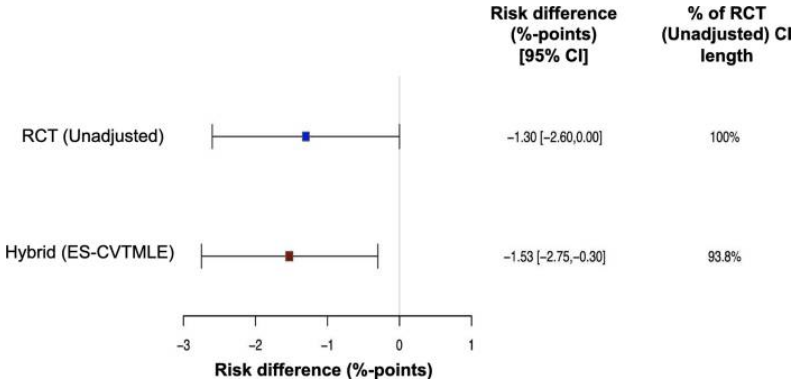
Motivating example: Oral semaglutide on MACE risk

- Scientific question: 1-year risk difference of MACE between **oral semaglutide** and **standard-of-care** group [3].
- Data: non-inferiority **RCT** + external controls from **Optum**.
- Results:



- Conclusion: Data integration has the potential to **enhance the efficiency** of RCT-only estimators, provide additional evidence for **testing superiority** within non-inferiority trials, thereby **streamlining the regulatory approval process**.

Challenges of augmenting RCTs with external RWD

Data challenges:

- **Inconsistencies** in key variable definitions/measurements between data sources:
  - Patients expected to follow treatment assignment in RCT, but in external RWD **treatment switches** may happen;
  - Systematic differences in **adherence** between controlled trial settings and routine clinical care;
- Non-concurrency in data collection (e.g., historical controls);
- Unmeasured confounding in the external data;

Statistical challenges:

- Efficiency gain is hard unless extra assumptions are imposed, i.e., efficient estimators would only use RCT data, hard to beat a semiparametrically efficient estimator (TMLE) applied to the RCT data [5].

Problem setup

We have access to both an RCT and RWD, each with treatment and control arms.

- $W \in \mathbb{R}^d$, patient baseline characteristics;
- $S \in \{0, 1\}$, study indicator ($S = 1$ for RCT, $S = 0$ for RWD);
- $A \in \{0, 1\}$, binary treatment;
- $Y \in \mathbb{R}$, outcome;
- Observe n i.i.d $O = (W, S, A, Y) \sim P_0 \in \mathcal{M}$.
- We factorize the density of O as follows

$$p(w, s, a, y) = p_W(w) p_S(s \mid w) p_A(a \mid s, w) p_Y(y \mid s, w, a).$$

- We have knowledge on $p_A(a \mid 1, w)$.

Goal: Achieve efficiency gain with minimal additional assumptions. Optimize power and precision while rigorously preserving valid inference for the causal estimand.

Candidate causal estimands

- ATE:

$$\tilde{\Psi}(P) = E(Y_1 - Y_0);$$

- ATE, averaged over pooled population:

$$\Psi(P) = E_W[E(Y_1 - Y_0 \mid S = 1, W)];$$

- ATE, averaged over RCT population:

$$\Psi_2(P) = E_{W|S=1}[E(Y_1 - Y_0 \mid S = 1, W)].$$

Comparing their identification assumptions

- **A1**. Mean ex. in the trial: $E(Y_a \mid S = 1, W, A = a) = E(Y_a \mid S = 1, W)$;
- **A2**. Pos. of receiving trt. in the trial: $0 < P(A = 1 \mid S = 1, W) < 1$;
- **A3**. Pos. of trial enrollment: $P(S = 1 \mid W) > 0$;
- **A4**. Mean ex. across the studies: $E(Y_a \mid S = 1, W) = E(Y_a \mid W)$.

	A1	**A2**	**A3**	**A4**
$\tilde{\Psi}$	✓	✓	✓	✓
Ψ	✓	✓	✓	
Ψ_2	✓	✓		

Challenges: Inconsistencies in outcome measurement between data sources, non-concurrency, unmeasured confounding in the external data, variations in the standard-of-care over time, and other differences between trial and routine care.

Choice of target estimand

- We consider

$$\Psi(P) = E_W E(Y_1 - Y_0 \mid S = 1, W)$$

  as our target estimand;

- Under identification assumptions, we have the statistical estimand

$$\Psi(P_0) = E_0[E_0(Y \mid S = 1, W, A = 1) - E_0(Y \mid S = 1, W, A = 0)].$$

But, ...

- One only needs $S = 1$ (RCT) data to estimate the inner expectations. External data is not being used!
- Can we utilize the external study $S = 0$ from the real world to obtain a more efficient estimator of the ATE, without adding assumptions?

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Targeted maximum likelihood estimation (TMLE)

- General statistical estimation framework for constructing **asymptotically linear** and locally **efficient plug-in** estimators of smooth target statistical estimands;
- Uses super-learner (ensemble learning with cross-validation selector) that leverages **state-of-the-art AI/ML** algorithms to obtain initial estimator/nuisance parameters;
- Targeting step that perturbs the initial estimates along a path through the initial likelihood estimator so that a unit increase in likelihood along this path corresponds with a maximal change in the target estimand, the targeting step **removes plug-in bias** and **enables valid statistical inference**;
- We need an estimator more efficient than a standard TMLE.

Adaptive TMLE

- A standard TMLE targeting step could push the initial estimator out of the working model in which the initial estimator lies in;
- Regularizing the targeting step by restricting it within a data-adaptively learned working model, estimating an oracle projection parameter instead of the nonparametrically defined target parameter;
- More regularized and therefore stable behavior in finite-sample, finite-sample super-efficiency, asymptotically valid inference due to oracle bias being second-order. The simpler the CATE is, the more efficiency gain one generally gets.

Adaptive TMLE general steps

1. Data-adaptively learn a working model $\mathcal{M}_n \subset \mathcal{M}$ with highly adaptive lasso [6, 7];
2. Define a projection parameter $\Psi_{\mathcal{M}_n} : \mathcal{M} \to \mathbb{R}$ as $\Psi_{\mathcal{M}_n}(P_0) = \Psi(\Pi_{\mathcal{M}_n} P_0)$ for some log-likelihood/loss-based projection of P_0 onto $\mathcal{M}_n$;
3. Construct a TMLE for $\Psi_{\mathcal{M}_n}(P_0)$ based on the canonical gradient of the projection parameter;
4. Provide Wald-type confidence intervals for $\Psi_{\mathcal{M}_n}(P_0)$ based on the canonical gradient of the projection parameter $D^{\star}_{\mathcal{M}_n, P_0}$.

A-TMLE applied to data integration setting

- Target estimand:

$$\Psi(P_0) = E_0[E_0(Y \mid S = 1, W, A = 1) - E_0(Y \mid S = 1, W, A = 0)]$$

- 1st component, **pooled-ATE** estimand:

$$\tilde{\Psi}(P_0) = E_0[E_0(Y \mid W, A = 1) - E_0(Y \mid W, A = 0)]$$

- 2nd component, **bias** estimand:

$$\Psi^{\#}(P_0) \equiv \tilde{\Psi}(P_0) - \Psi(P_0)$$
$$\Psi(P_0) = \tilde{\Psi}(P_0) - \Psi^{\#}(P_0) \text{ bias correction}$$

- We apply A-TMLE to estimate $\tilde{\Psi}$ and $\Psi^{\#}$, respectively.

Why consider this decomposition?

- With some algebra, the bias estimand could be written as

$$\Psi^{\#}(P_0) = E[\Pi(0 \mid W, 0)\tau_S(W, 0) - \Pi(0 \mid W, 1)\tau_S(W, 1)],$$

where

$$\Pi(s \mid W, A) = P(S = s \mid W, A),$$

and

$$\tau_S(W, A) = E(Y \mid S = 1, W, A) - E(Y \mid S = 0, W, A).$$

- **Key insight:** the amount of efficiency gain from A-TMLE has to do with the **complexity** of the working model;
- **The bias function might be simple**;
- Therefore, by exploiting the potentially simple and learnable structure of the bias working model, we might have efficiency gain compared to TMLE.

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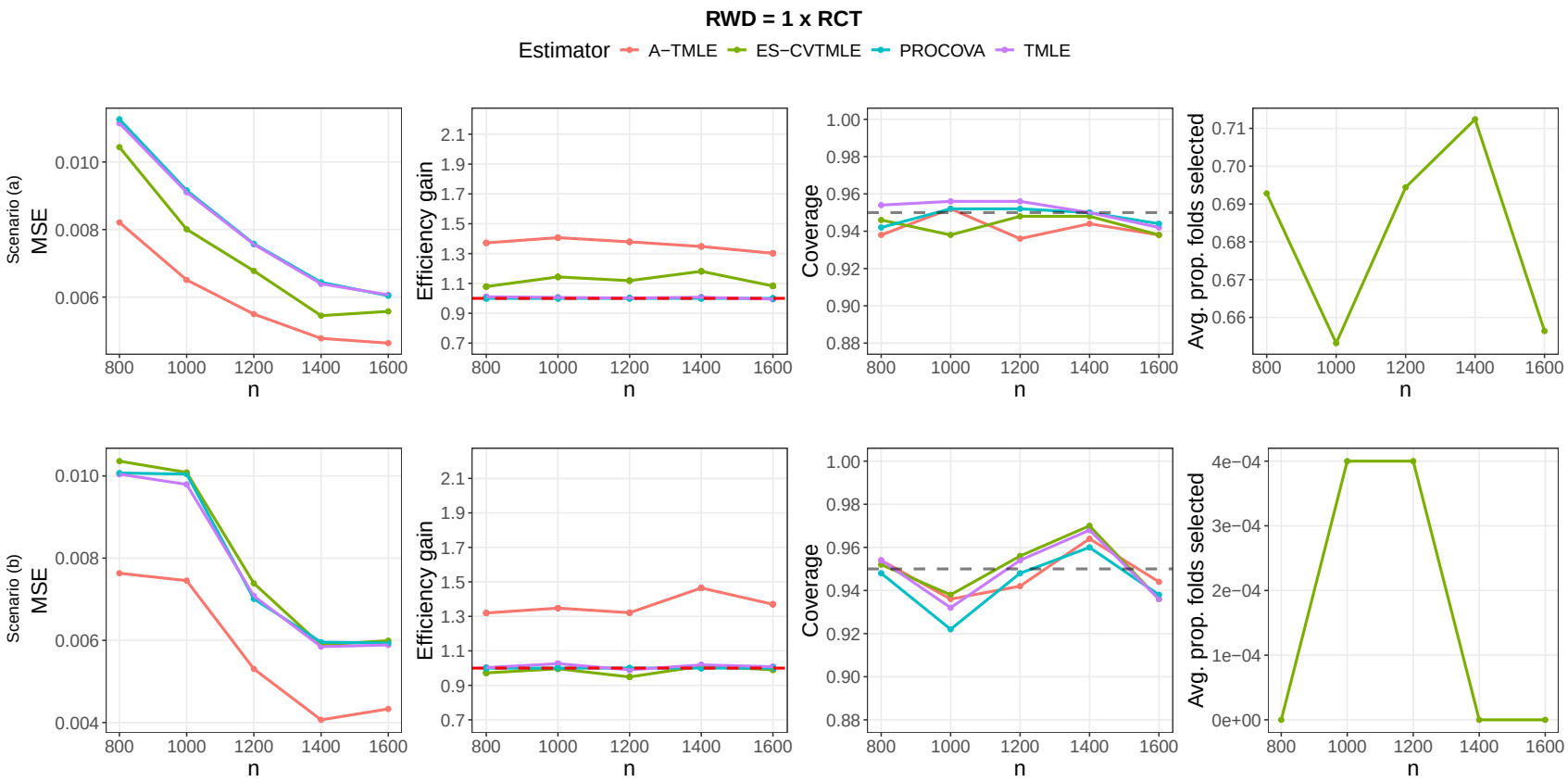
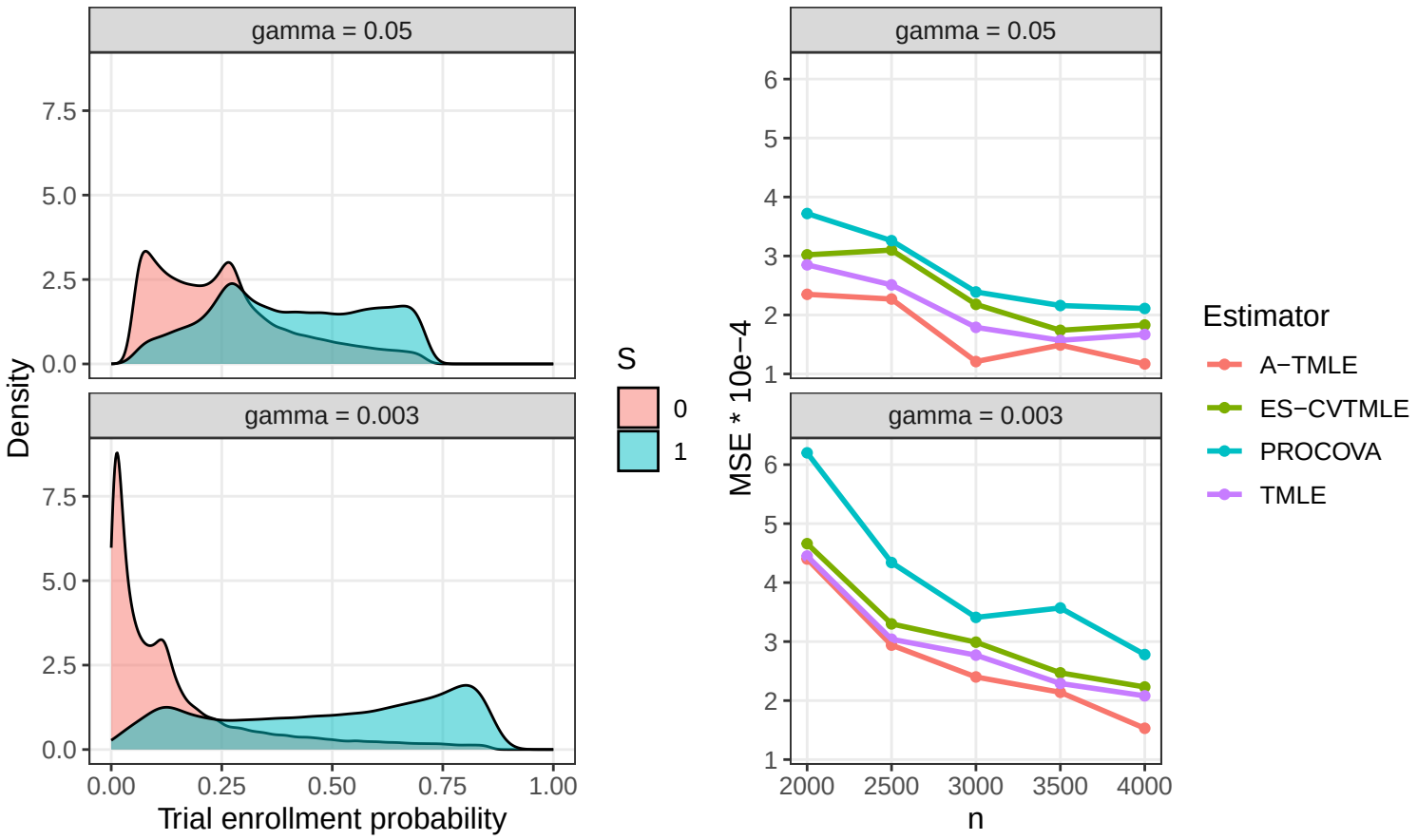
Simulations: operating characteristics



Simulations: severe trial enrollment positivity violations



γ is the infimum of trial enrollment probability.

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Case study: DEVOTE + Optum

- DEVOTE: evaluate the efficacy and safety of insulin degludec versus glargine in type 2 diabetes [8];
- Treat-to-target, randomized, double-blind, active comparator-controlled, event-driven, non-inferiority;
- **Treatment:** insulin degludec; **Comparator:** insulin glargine;
- **Primary outcome:** Composite, first occurrence of an major cardiovascular event (death from cardiovascular causes, nonfatal myocardial infarction, or nonfatal stroke);
- **Scientific question:** 1.5- and 2-year risk differences of MACE between the treatment and comparator arm;
- **External data source:** Optum's de-identified Clinformatics® Data Mart Database, has both degludec treated and glargine treated patients. From 2013 - 2021.

How to select external patients?

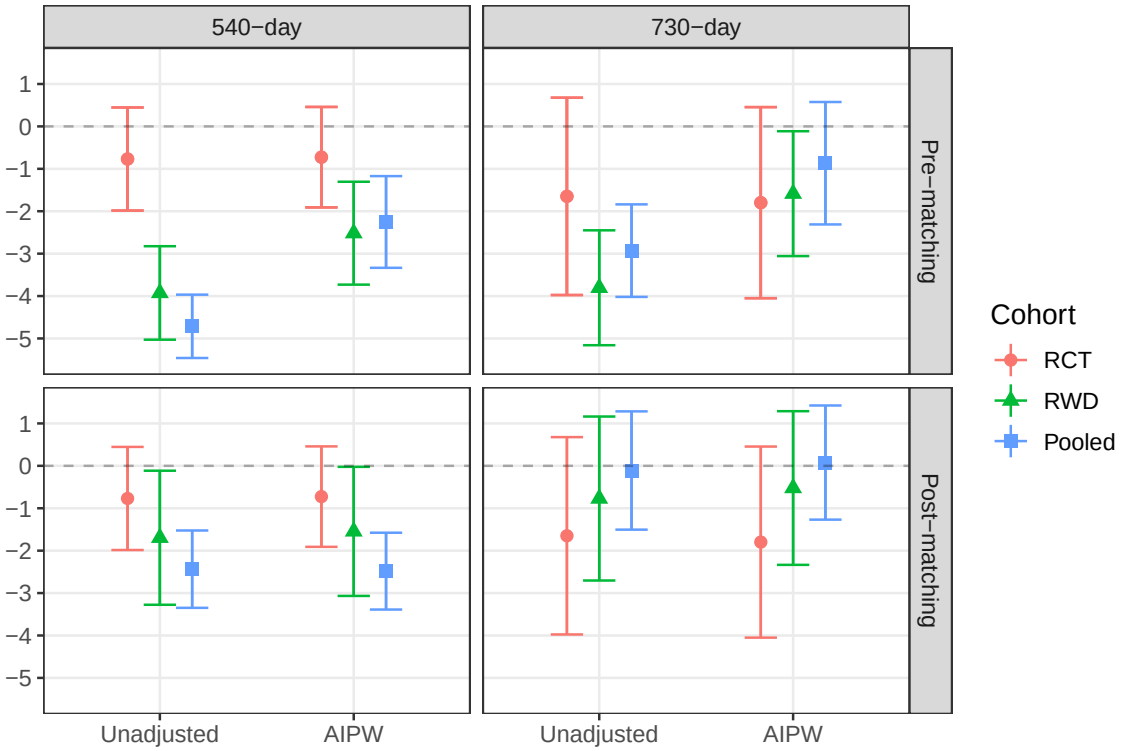
- Why not just use all the external data available?
- How to obtain a subsample from the external RWD to improve the robustness of the estimator and efficiency gain?
- Preferably outcome-blind.

Theorem

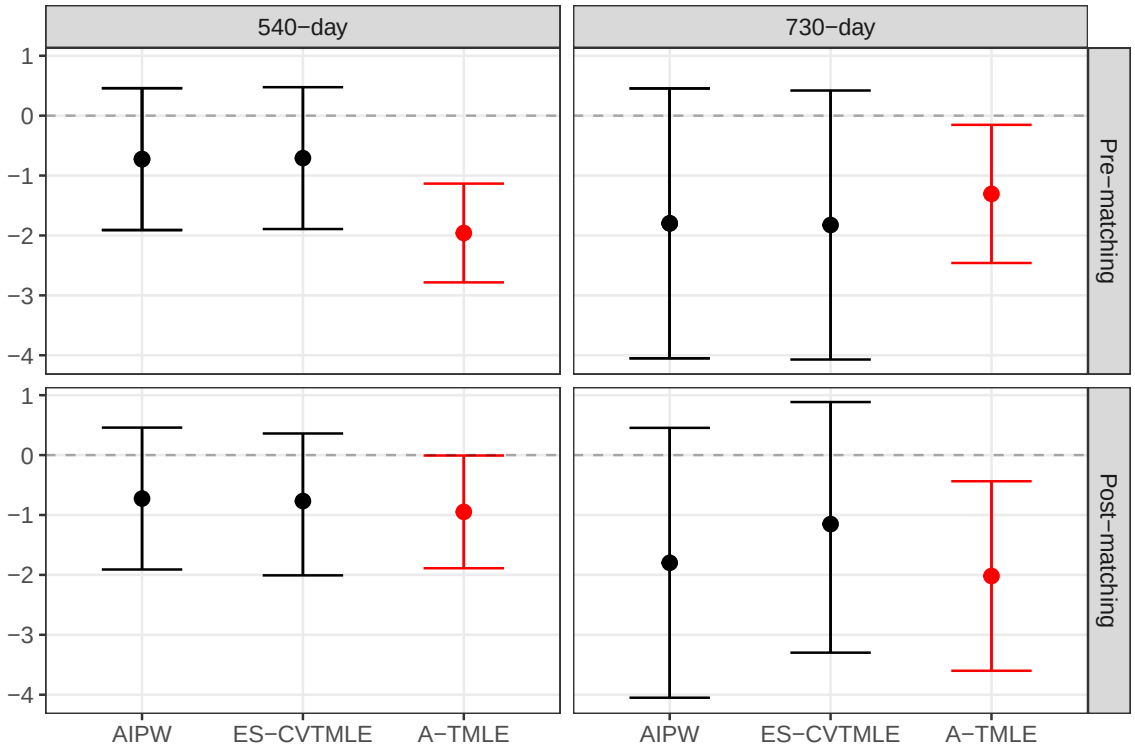
If the trial enrollment score $P_0(S = 1 \mid W)$ and the propensity score in the external data $P_0(A = 1 \mid S = 0, W)$ are correctly specified, then the exact remainder $R_{\mathcal{M}_w}(P, P_0) \equiv \Psi_{\mathcal{M}_w}(P) - \Psi_{\mathcal{M}_w}(P_0) + P_0 D_{\mathcal{M}_w, P} = 0$. Further, if those two scores are constant in W, then the oracle bias $\Psi_{\mathcal{M}_w}(P_0) - \Psi(P_0) = 0$. In the special case of augmenting with external controls only (i.e., no external treatment arm), $P_0(A = 1 \mid S = 0, W) = 0$, which is already a constant. Therefore, in this setting, the only condition required for both the exact remainder for the projection estimand and the oracle bias to be zero is that $P_0(S = 1 \mid W)$ must also be constant in W.

Selecting external patients

- 1:3 trial enrollment score matching, 1:5 external treated to control propensity score matching;
- **RCT:** trt. 3818, con. 3817; **RWD:** trt. 2359, con. 11795.



Data integration analysis



Data integration analysis results. ES-CVTMLE and A-TMLE point estimates with corresponding 95% confidence intervals for the 540-day and 730-day risk difference (%) of MACE, both pre- and post-matching. AIPW estimates are based on RCT data only.

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Software

- Support for Ψ_2
- Replace relaxed targeting with TMLE type targeting
- Data fusion benchmark suite

Concluding remarks

- We introduced an estimator that allows **more efficient** estimation of ATE in the RCT by leveraging external RWD with **minimal additional assumptions**;
- We demonstrated in a case study that our proposed estimator successfully **improved the power** of the DEVOTE non-inferiority trial when augmented with external data and provided statistically significant results for **testing for treatment effect superiority**;
- Opportunities for using these data integration estimators for **subgroup analysis**, **transportation** studies;
- Utilizing RWD to improve the efficiency of RCTs could allow for **smaller, faster trials, decreasing the time** until patients can receive effective interventions.

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